

# Solution for CS5275 Problem Set 2

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# 1 Problem 1

Consider the following procedure. Starting from an  $n$ -vertex graph  $G = (V, E)$ , iteratively do the following: as long as there exists a component  $C$  and a cut  $(S, C \setminus S)$  such that  $\Phi_C(S) \leq \phi$ , erase the edges in the cut, thus splitting  $C$  into two parts (each of one or more components). The procedure stops when there is no more  $S$  that can be found.

Prove that the total number of edges erased is

$$O(\phi \cdot |E| \cdot \log n).$$

## 1.1 My solution

In this solution, we will use a slight abuse of notation. If  $A, B$  are subsets of the vertex set  $V$ , then  $\text{vol}_A(B) = \text{vol}_{G[A]}(B)$  is the volume of  $B$  in the induced subgraph  $G[A]$ .

We append the following to the procedure. We are going to put “pebbles” on vertices. A vertex can have any number of pebbles, and at the beginning, there is no pebble anywhere. The “weight” of a pebble on vertex  $v$  is  $w(v) := \text{vol}_G(\{v\}) = \deg_G(v)$ .

Whenever we make a cut  $(S, C \setminus S)$ , put a pebble on every vertex on the smaller side. In other words, suppose  $\text{vol}_C(S) \leq \text{vol}_C(C \setminus S)$  – otherwise we take  $S \leftarrow C \setminus S$  – then put a pebble on every vertex in  $S$ .

We make the following two claims:

**Lemma 1.1.** *Consider any step of the procedure, suppose it makes the cut  $(S, C \setminus S)$ , and suppose  $S$  is the smaller side. Then*

$$|E(S, C \setminus S)| \leq \phi \cdot \sum_{v \in S} w(v).$$

*In other words, the number of edges cut is at most  $\phi$  times the total weight of pebbles put with this cut.*

**Lemma 1.2.** *At the end of the procedure, each vertex has  $\leq 2 \log_2 n$  pebbles.*

We first prove these two lemmas, then complete the proof using these lemmas.

*Proof of Lemma 1.1.* This is straightforward from the definition of conductance  $\Phi$ . By definition of the procedure, we have

$$\Phi_C(S) \leq \phi \quad \implies \quad \frac{|E(S, C \setminus S)|}{\min\{\text{vol}_C(S), \text{vol}_C(C \setminus S)\}} \leq \phi$$

Since  $S$  is the smaller side,

$$|E(S, C \setminus S)| \leq \phi \cdot \text{vol}_C(S).$$

Finally,

$$\text{vol}_C(S) \leq \text{vol}_G(S) = \sum_{v \in S} \deg_G(v) = \sum_{v \in S} w(v)$$

This proves the lemma. □

*Proof of Lemma 1.2.* We claim the following. Suppose a vertex  $v$  is in a component  $C$ . At some step,  $C$  is cut so that  $v$  now belongs in a smaller component  $S$ . Then if  $v$  gets a pebble, then

$$\text{vol}_S(S) \leq \frac{\text{vol}_C(C)}{2}.$$

In other words, whenever a vertex gets pebbled, its component has been halved.

But this is straightforward. We have  $\text{vol}_S(S) \leq \text{vol}_C(S)$ , and if  $S$  is pebbled, that means  $\text{vol}_C(S) \leq \text{vol}_C(C \setminus S)$ . Therefore  $\text{vol}_S(S) \leq \text{vol}_C(S) \leq \text{vol}_C(C)/2$ .

Now,  $\text{vol}_V(V) \leq n^2$  since  $G$  is unweighted. Therefore a vertex  $v$  can only be pebbled up to  $2 \log_2 n$  times before its component has volume  $< 1$ . When this happens, the component has volume 0, i.e.  $v$  is an isolated vertex. Then it will not be involved in any future cut. This proves the claim. □

We now return to the main proof. Suppose the procedure has finished. Let  $E^\times$  be the set of removed edges, and define  $p_v$  to be the number of pebbles on vertex  $v$ . By Lemma 1.1 summed over all cuts,

$$|E^\times| \leq \phi \cdot \sum_v (p_v \cdot w(v)).$$

By Lemma 1.2,  $p_v \leq 2 \log n$ . Also recall  $w(v) = \deg_G(v) = \text{vol}_G(\{v\})$ . Therefore

$$\begin{aligned} |E^\times| &\leq \phi \cdot \sum_v (p_v \cdot w(v)) \\ &\leq \phi \cdot 2 \log_2 n \cdot \sum_v w(v) \\ &= \phi \cdot 2 \log_2 n \cdot \text{vol}_G(V) \\ &= \phi \cdot 2 \log_2 n \cdot 2 \cdot |E| \\ &= O(\phi \cdot |E| \cdot \log n) \end{aligned}$$

## 1.2 Insight and discussion

The core result is Lemma 1.2: a vertex only “contributes” to a cut at most  $O(\log n)$  times. The insight is pretty straightforward when spelled out: the main contribution to a cut size comes from the volume of the smaller side, so whenever a vertex contributes to a cut, its side has been halved. The graph itself only has  $O(n^2)$  volume, so you can only halve  $O(\log n)$  times before the volume becomes  $< 1$ .

The phrasing of the proof by using pebbles is how the solution literally came to my mind. There are several different variations on how this concept is written in your solutions. Some look for how many times a vertex “contributes” to the cut. Some use “charge”, as in the cost of the cut is charged to each vertex. (In this case, the charge for  $v$  is  $\phi \cdot \deg(v)$ , and  $|E(S, C \setminus S)| \leq \text{sum of charges}$ .) They are all basically the same, and they all work.

## 1.3 Grading criteria and common mistakes

### 1.3.1 Major error: Counting number of cuts

To obtain the  $O(\log n)$  bound, you need to count the number of cuts that a vertex “contributes”. Not just the overall total number of cuts, because there can be  $O(n)$  of them: consider taking away 1 vertex each cut.

Perhaps you want to arrange the cuts as a tree. Cutting a component splits a node into two (or more) children corresponding to the components of the cut. Now, cuts made on disjoint components are always disjoint, so you might try to count the number of levels of the tree. The problem is, the same counterexample above shows that the tree might have  $O(n)$  levels.

So if you simply count the number of cuts, it will not work, your bound is too loose. This is a major part of the problem, so it counts as a major error.

If you have some sort of  $O(\log n)$  calculation (using the fact that you halve a component’s volume), but fail to link it correctly to your proof, it’s a minor error, and your solution will likely get marked 3 points of 4.

## 1.4 Another solution: Grouping cuts

As explained in Section 1.3.1, you need to derive the  $O(\log n)$  bound in a smart way, you cannot simply count cuts. Even trying to form a tree is tricky. Most students prove the  $O(\log n)$  bound using the pebble argument in the main solution. But there is a solution that sort of represents the tree argument to derive the same  $O(\log n)$  bound.

We will group the cuts into rounds, where in each round, the smaller parts of the cuts are vertex-disjoint. I want to emphasize what parts are vertex-disjoint here: only the *smaller* parts of the cuts are vertex-disjoint. It is possible there’s a cut of  $C$  into  $(X, Y)$ , and then  $Y$  into  $(A, B)$ , both in the same round, as long as the smaller part of  $(X, Y)$  is  $X$  instead of  $Y$ . If we represent each component as a node on a tree, we’re not quite

counting the number of levels; we have some sort of “slanted” levels that may group nodes from different levels together.

First, note that many of the cuts are commutative; as long as they act on disjoint components, the cuts can be done in any order. So, suppose the procedure runs and creates a series of cuts. At any given point during the procedure, some of the cuts have been made, and some have not yet been made. The commutativity means, we may pick any of the remaining cuts that is currently possible. The ordering doesn’t matter; doing all the cuts will lead to the same set of components at the end.

We will now group the cuts into **rounds**. In each round, if a component ends up as the smaller side, it will not be cut further this round.

Components will be labeled by one of three terms: **thawed**, **frozen**, or **dead**. A thawed component may be involved in a cut this round. A frozen component can no longer be cut this round, but will thaw next round. A dead component will never be cut any more.

At the beginning of the entire procedure, each component of  $G$  is thawed.

In each round, we will only cut thawed components. Whenever we cut a thawed component  $C$  into a smaller side  $S$  and a larger side  $C \setminus S$ , we say  $S$  becomes frozen and  $C \setminus S$  remains thawed.

At the end of each round, any remaining thawed component becomes dead. They were eligible to cut, but they are not cut, meaning they will never be involved in any future cut. After that, all frozen components become thawed for the next round.

The procedure continues until all components become dead. We will now put a bound on the number of rounds.

Suppose we have a cut  $(S, C \setminus S)$  where  $S$  is frozen. That means  $|E(S, C \setminus S)| \leq \phi \cdot \text{vol}_C(S) \leq \phi \cdot \text{vol}_G(S)$  by the same argument as in the main solution. Put another way, a cut is paid for by freezing the smaller side; freezing a component  $S$  pays for a cut of size  $\leq \phi \cdot \text{vol}_G(S)$ .

Now consider a single round. Suppose the frozen components are  $C_1, C_2, \dots, C_k$ . Because the frozen components are all vertex-disjoint, the total number of edges cut this round is

$$\leq \phi \cdot \sum_i \text{vol}_G(C_i) \leq \phi \cdot \text{vol}_G(V) = \phi \cdot 2|E|.$$

Now consider the following observation. Suppose at the beginning of a round, all non-dead components have volume  $\leq K$ . Then we claim, at the end of a round, all non-dead components have volume  $\leq K/2$ . But this is straightforward. The only way a component  $S$  is not dead at the end of a round is if it was frozen during the round, and then thawed after there are no more cuts to be made this round. If  $S$  was frozen, that means it was the smaller part of a cut  $(S, C \setminus S)$ . But  $\text{vol}_G(C) \leq K$ , so  $\text{vol}_G(S) \leq K/2$ .

Thus by induction, we can conclude: after the  $k$ -th round, all non-dead components have volume  $\leq \text{vol}_G(V) \cdot 2^{-k}$ . Since  $\text{vol}_G(V) \leq n^2$ , after  $k = 2 \log n + 1$  rounds, all non-dead components have volume  $\leq 1/2$ . But volume is an integer, so the volume must be zero. That means each non-dead component is an isolated vertex, and it will become dead in the  $(2 \log n + 2)$ -th round.

Therefore, there are at most  $2 \log n + 2$  rounds. In each round, we cut  $\leq \phi \cdot 2|E|$  edges total. The conclusion follows.

## 1.5 Another solution: Potential function

A student uses a “potential” function, inspired by the potential method used in analysis of algorithms. Normally we use  $\Phi$  for the potential, but because that symbol is already being used, we’ll use  $P$  instead.

Let  $C$  be a component. Then

$$P(C) = \phi \cdot \text{vol}_C(C) \cdot \log_2(\text{vol}_C(C) + 1).$$

(All logarithm in this solution will be base 2. The  $+1$  is so the potential doesn’t vomit out on an isolated vertex with volume 0.)

For a set of components  $X$ , the potential is just the sum of the potentials of the individual components; i.e.  $P(X) = \sum_{C \in X} P(C)$ . Note that  $P$  is a convex function on  $|C|$ , so  $P(X) \leq \phi \cdot \text{vol}_X(X) \cdot \log_2(\text{vol}_X(X) + 1)$ .

Now, if we cut a component  $C$  into  $S$  and  $C \setminus S$ , consider the change of potential. Before the cut, the potential is  $P(C)$ . After the cut, the potential is  $P(S) + P(C \setminus S)$ . We now give a bound on the difference:

$$\begin{aligned} P(C) - (P(S) + P(C \setminus S)) \\ &\geq \phi \cdot \text{vol}_C(C) \log(\text{vol}_C(C) + 1) - (\phi \cdot \text{vol}_S(S) \log(\text{vol}_S(S) + 1) + \phi \cdot \text{vol}_{C \setminus S}(C \setminus S) \log(\text{vol}_{C \setminus S}(C \setminus S) + 1)) \\ &\geq \phi \cdot (\text{vol}_C(C) \log(\text{vol}_C(C) + 1) - \text{vol}_C(S) \log(\text{vol}_S(S) + 1) - \text{vol}_C(C \setminus S) \log(\text{vol}_C(C \setminus S) + 1)) \end{aligned}$$

(The first inequality is an inequality because  $S$  and  $C \setminus S$  might each be split further into components. This is why I mentioned  $P$  is convex. In the second inequality, we weaken  $\text{vol}_{C \setminus S}(C \setminus S) \leq \text{vol}_C(C \setminus S)$ , but don't touch  $\text{vol}_S(S)$  for reasons to be evident soon.)

Now, we claim that

$$\text{vol}_S(S) + 1 \leq \frac{1}{2} \cdot (\text{vol}_C(C) + 1).$$

In particular, this allows us to take the logarithm:

$$\log(\text{vol}_S(S) + 1) \leq \log(\text{vol}_C(C) + 1) - 1.$$

The proof is simple. We know  $\text{vol}_C(S) \leq \text{vol}_C(C)/2$  since it's the smaller side. In addition, we said  $C$  is a component, so the cut removes at least 1 edge. So  $\text{vol}_S(S) \leq \text{vol}_C(S) - 1$ . This gives the claim.

We apply this to the second term above to extract out a  $\phi \cdot \text{vol}_C(S)$  component:

$$\begin{aligned} P(C) - (P(S) + P(C \setminus S)) \\ &\geq \phi \cdot \left( \text{vol}_C(C) \log(\text{vol}_C(C) + 1) - \text{vol}_C(S) \cdot [\log(\text{vol}_C(C) + 1) - 1] - \text{vol}_C(C \setminus S) \log(\text{vol}_C(C) + 1) \right) \\ &= \phi \cdot \left( \log(\text{vol}_C(C) + 1) \cdot (\text{vol}_C(C) - \text{vol}_C(S) - \text{vol}_C(C \setminus S)) + \text{vol}_C(S) \cdot 1 \right) \\ &= \phi \cdot \text{vol}_C(S) \end{aligned}$$

Therefore the difference of the potential is  $\geq \phi \cdot \text{vol}_C(S)$ . By the conductance, this is  $\geq |E(S, C \setminus S)|$ , so the potential lost covers the number of edges cut.

At the beginning, the potential is  $\phi \cdot \text{vol}_G(V) \log(\text{vol}_G(V) + 1)$ . But we know  $\text{vol}_G(V) = |E| \leq n^2$ , so  $\log(\text{vol}_G(V) + 1) \leq 2 \log n + 1 = O(\log n)$ . At the end, every volume is  $\geq 0$ , so the potential is  $\geq 0$ . Therefore the total number of edges removed is at most the potential:

$$\phi \cdot \text{vol}_G(V) \log(\text{vol}_G(V) + 1) = O(\phi \cdot |E| \cdot \log n).$$

## 2 Problem 2

In this problem, let  $G$  be a graph with normalized Laplacian  $\mathbf{N}$ .

- (a) For every  $n$ , construct a connected graph  $G$  with  $n$  vertices, such that

$$\lambda_2(\mathbf{N}) = O(1/n^3).$$

- (b) Prove that any connected graph  $G$  on  $n$  vertices satisfies

$$\lambda_2(\mathbf{N}) = \Omega(1/n^3).$$

### 2.1 My solution

For simplicity, I will use  $\lambda_2$  to mean  $\lambda_2(\mathbf{N})$ . Fix a graph  $G = (V, E)$ , and let  $f : V \rightarrow \mathbb{R}$ . Write

$$\lambda_2(f) = \frac{\sum_{uv \in E} (f(u) - f(v))^2}{\sum_v \deg(v) f(v)^2}$$

Then  $\lambda_2 = \min_f \lambda_2(f)$ , among all  $f$  satisfying  $\sum_v \deg(v) f(v) = 0$  and  $f \not\equiv 0$ .

**Part (a).** We will use the “thin bridge” construction (previously used in Problem set 1, Problem 4, as an appendix). For simplicity, let  $n = 4k - 1$ ; read the end of the proof to adapt this construction for other  $n$ . The graph  $G$  has the following form:

Take  $k$  vertices into a clique, and call it the “positive island”  $C_+$ . Take  $k$  more vertices into another clique, and call it the “negative island”  $C_-$ . Then take an arbitrary vertex in  $C_+$  and label it  $v_{+k}$ , and take another arbitrary vertex in  $C_-$  and label it  $v_{-k}$ . Finally, connect  $v_{+k}$  and  $v_{-k}$  with a path of length  $2k$  edges, with vertices  $v_{-k}, v_{-k+1}, v_{-k+2}, \dots, v_{+k}$ .

We also define the following function  $f$ :

$$f(v) = \begin{cases} i & \text{if } v = v_i \text{ is on the thin bridge} \\ k & \text{if } v \text{ is on the positive island } C_+ \\ -k & \text{if } v \text{ is on the negative island } C_- \end{cases}$$

We claim this  $f$  satisfies the conditions and so counts for  $\lambda_2$ , and  $\lambda_2(f) = O(k^{-3}) = O(n^{-3})$ . Therefore  $\lambda_2 \leq \lambda_2(f) = O(n^{-3})$ , proving the problem statement.

To see that  $\sum_v \deg(v) f(v) = 0$ , just note that the graph is symmetrical around  $v_0$ , and  $f$  is anti-symmetrical around it. So this  $f$  counts for  $\lambda_2$ . Now we compute the numerator and the denominator of  $\lambda_2(f)$ .

**Numerator.** Edges  $uv$  in either clique have  $f(u) - f(v) = 0$ . The remaining edges are the ones along the thin bridge, which have  $(f(u) - f(v))^2 = 1$ . There are  $2k$  such edges, so the numerator is  $2k$ .

**Denominator.** Consider the positive island  $C_+$  by itself. There are  $k$  vertices here. Each vertex here has degree  $\Omega(k)$  (specifically,  $k - 1$ , except for  $v_{+k}$  which has degree  $k$ ). Each vertex here has  $f = k$ . So the positive island contributes

$$\sum_{v \in C_+} \deg(v) f(v)^2 = k \cdot \Omega(k) \cdot k^2 = \Omega(k^4)$$

to the denominator. Therefore the denominator is  $\Omega(k^4)$ . (Indeed, the negative island also contributes  $\Omega(k^4)$ ; the thin bridge only contributes  $\Omega(k^3)$ .)

Therefore

$$\lambda_2 \leq \lambda_2(f) = \frac{2k}{\Omega(k^4)} = O(k^{-3}) = O(n^{-3}).$$

**Handling other  $n$ .** The above construction works for  $n = 4k - 1$ . To make it work for other  $n$ , simply attach the additional vertices to the center with  $f = 0$ . This doesn't change any of the calculations, and the asymptotic is still correct since  $k = \Theta(n)$  still.

**Part (b).** Let  $f$  be any function satisfying the conditions of the problem. We will show  $\lambda_2(f) = \Omega(n^{-3})$ . For this problem, recall that  $G$  is connected, and we assume  $G$  has  $\geq 2$  vertices.

First, let  $M = \max |f(v)|$  be the maximum absolute value of  $f$ . Without loss of generality, suppose  $M$  is achieved positively, i.e. there is some  $v_{\max}$  such that  $f(v_{\max}) = M$ , otherwise negate all values of  $f$ .

Furthermore, consider the condition  $\sum_v \deg(v)f(v) = 0$ . Now  $\deg(v) > 0$  for all  $v$ , so there must be some  $v_{\min}$  for which  $f(v_{\min}) \leq 0$ , otherwise the sum is strictly positive.

Now we look at the numerator and the denominator of  $\lambda_2(f)$ .

**Numerator.** There is a path from  $v_{\min}$  to  $v_{\max}$  because  $G$  is connected; label the vertices of this path as  $v_0, v_1, v_2, \dots, v_k$  where  $v_0 = v_{\min}$  and  $v_k = v_{\max}$ . Note that  $k \leq n$  because there are only  $n$  vertices in the graph.

We consider the contribution of this path to the numerator. Using the [QM-AM inequality](#)<sup>1</sup>:

$$\frac{\sum_{i=0}^{k-1} (f(v_i) - f(v_{i+1}))^2}{k} \geq \left( \frac{\sum_{i=0}^{k-1} |f(v_i) - f(v_{i+1})|}{k} \right)^2$$

Then by using the triangle inequality:

$$\sum_{i=0}^{k-1} |f(v_i) - f(v_{i+1})| \geq |f(v_k) - f(v_0)| = |f(v_{\max}) - f(v_{\min})| \geq M$$

Therefore, the numerator is

$$\geq \sum_{i=0}^{k-1} (f(v_i) - f(v_{i+1}))^2 \geq \frac{1}{k} \cdot \left( \sum_{i=0}^{k-1} |f(v_i) - f(v_{i+1})| \right)^2 \geq \frac{1}{n} \cdot M^2 = \Omega(M^2 n^{-1})$$

**Denominator.** There are  $n$  vertices, each of which has  $\deg(v) = O(n)$ . Moreover, by definition of  $M$ , we have  $f(v)^2 \leq M^2$ . Therefore, the denominator is

$$\sum_v \deg(v) f(v)^2 \leq n \cdot O(n) \cdot O(M^2) = O(M^2 n^2)$$

Therefore,

$$\lambda_2(f) = \frac{\Omega(M^2 n^{-1})}{O(M^2 n^2)} = \Omega(n^{-3}).$$

This is true for all  $f$ . Therefore

$$\lambda_2 = \min_f \lambda_2(f) = \Omega(n^{-3}).$$

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<sup>1</sup>also Cauchy-Schwarz

## 2.2 Insight and discussion

Both parts of the problem stem from the same observation: in order for  $\lambda_2$  to have a low result, we want  $(f(u) - f(v))^2$  to be small all the time, but  $f(v)^2$  to be big. The problem is that these two objectives are opposing each other.

The main observation is the QM-AM inequality. You can motivate it from the following. The condition requires there to be some  $v_{\min}$  with  $f(v_{\min}) \leq 0$ . Now you have a  $v_{\max}$  with large (positive)  $f(v_{\max})$ . You have a path from  $v_{\min}$  to  $v_{\max}$ , so intuitively, the best way to keep  $\sum (f(u) - f(v))^2$  low is by making gradual incremental steps; each step isn't too large by itself, but the sum over all steps gets you all the way from  $f(v_{\min})$  to  $f(v_{\max})$ . And to formalize this, you can use the QM-AM inequality (or Cauchy-Schwarz). If you start with this observation, you will likely solve part (b) first.

Now you know, for part (a), you need all the inequalities to be tight. For example, the numerator is  $\Omega(M^2 n^{-1})$ , so you need this to be actually  $\Theta(M^2 n^{-1})$ . This can happen by having a  $\Theta(n)$ -long path that gets you from 0 to  $M$ . On the other hand, the denominator is  $O(M^2 n^2)$ . So you need this to be  $\Theta(M^2 n^2)$ , by having  $\Theta(n)$  vertices at degree  $\Theta(n)$  that all have  $f = M$ . This can happen by having a  $\Theta(n)$ -big clique with  $f = M$ . And piecing this together gives you something like the thin bridge construction.

So, while the construction isn't very natural to come by, writing down all the constraints you have to satisfy can lead you there.

## 2.3 Grading criteria and common mistakes

### 2.3.1 Fatal error: Using Cheeger's inequality

Cheeger's inequality says,

$$\frac{\lambda_2}{2} \leq \Phi(G) \leq \sqrt{2\lambda_2}$$

But this will not help you with anything. You know  $\Phi(G) \geq 1/|E| = \Omega(n^{-2})$  from lecture 1 slide 8, and you can indeed have  $\Phi(G) = O(n^{-2})$  by e.g. a barbell graph (two cliques joined by a single edge). But this just means  $\lambda_2 = \Omega(n^{-4})$  in general, and  $\lambda_2 = O(n^{-2})$  for the barbell graph and other examples. Neither gets you to the  $n^{-3}$  bound you want.

In general, any approach using Cheeger's inequality will not work, and will not get any credit.

For (a), one student claimed that the  $O(n^{-2})$  bound is for  $\lambda_2(\mathbf{A})$  with the adjacency matrix  $\mathbf{A}$ , so since  $\mathbf{N}$  is obtained by scaling by  $1/n$ , it follows  $\lambda_2(\mathbf{N}) = O(n^{-3})$ . This is wrong; Cheeger's inequality already applies for  $\lambda_2(\mathbf{N})$ .

### 2.3.2 (a) Major error: Single edge instead of long path

If you use a single edge instead of a long path between the two cliques, your construction doesn't work. The numerator just has a single edge, but it has  $(f(u) - f(v))^2 = (k - (-k))^2 = 4k^2 = \Theta(k^2)$ . This causes  $\lambda_2(f) = \Omega(k^{-2})$ . You need a long path to "smoothen" out the descent from one clique to the other, so the numerator remains small.

### 2.3.3 (a) Passable error: Not defining "barbell graph"

Some of you don't explain what a "barbell graph" or a "dumbbell graph" is. According to [Wolfram MathWorld](#), there are several definitions of a barbell graph; the more common definition is that two cliques are joined by only a single edge rather than a long path. This would lead to the major error in Section 2.3.2.

If you also state something like, we linearly interpolate  $f$  along the "path", then I assume you're using the correct construction and give full credit. Just remember to be careful with your definitions, especially when there are multiple different ones.



### 2.3.4 (b) Minor error: Not showing the range is $\geq M$

When computing the numerator, we use the inequality  $|f(v_{\max}) - f(v_{\min})| \geq M$ , which comes from  $\sum_v \deg(v)f(v) = 0$ . This is an important observation. Without making this observation, it's possible that  $f$  is constant nonzero, and then  $|f(v_{\max}) - f(v_{\min})| = 0$ . This breaks the analysis.

It's a pretty simple observation, so I don't want to deduct too much for this. But it's an important one, and I cannot give full credit if you're missing this observation.

### 2.3.5 (b) Passable error: Only using $\max f$

In the main solution,  $M = \max_v |f(v)|$ ; note the absolute value. Some students simply take  $M = \max_v f(v)$ , and still bound the denominator by  $O(M^2 n^2)$ .

This is not correct in general, because it's possible  $m = \min_v f(v)$  satisfies  $|m| > |M|$ . Then the denominator is actually  $O(m^2 n^2)$  instead, which is a larger number, pushing  $\lambda_2$  smaller.

This is simple to fix though, so I do not deduct points for this. But it's still an error, and you should be careful about this.

### 2.3.6 (b) Passable error: Minimizing numerator is not formal

Some of you argue the numerator is minimized when the steps  $|f(v_i) - f(v_{i+1})|$  are all as small as possible, so they have to be evenly distributed.

This is the intuition, but it is not a formal proof. You can prove this formally by using QM-AM inequality. But I think it's minor enough that I don't deduct anything for this.

## 2.4 (b) Another solution: Different manipulation of the denominator

I found several solutions that manipulate the denominator in a different way, to obtain some term in the form  $\sum_{uv \in E} (f(u) - f(v))^2$ , so it can cancel the numerator.

The primary expression we'll be investigating is the following:

$$\sum_u \sum_v \deg(u) \deg(v) \cdot (f(u) - f(v))^2$$

It's a bit difficult to come up with this expression, but the main motivations are as follows:

- It clearly has a  $(f(u) - f(v))^2$  term, which we will bound by  $\leq (f(v_{\max}) - f(v_{\min}))^2$ .
- When we break down the sum, we have terms in the form  $\sum_u \deg(u)f(u)^2$ , so the denominator can be written as this expression in some way.
- Also, we have terms in the form  $\sum_u \deg(u)f(u)$ , which is 0 by the condition.
- We also have  $\sum_u \deg(u) = 2 \cdot \text{vol}(G) \leq n^2$ , which will give rise to all the  $n$ 's in  $\Omega(n^{-3})$ .

Now we will work out the expression:

$$\begin{aligned} & \sum_u \sum_v \deg(u) \deg(v) \cdot (f(u) - f(v))^2 \\ &= \sum_u \sum_v \deg(u) \deg(v) \cdot f(u)^2 - 2 \cdot \sum_u \sum_v \deg(u) \deg(v) \cdot f(u)f(v) + \sum_u \sum_v \deg(u) \deg(v) \cdot f(v)^2 \\ &= \sum_v \deg(v) \cdot \left( \sum_u \deg(u)f(u)^2 \right) - 2 \cdot \left( \sum_u \deg(u)f(u) \right) \cdot \left( \sum_v \deg(v)f(v) \right) + \sum_u \deg(u) \cdot \left( \sum_v \deg(v)f(v)^2 \right) \\ &= 2 \cdot \text{vol}(G) \cdot (\text{denominator}) - 2 \cdot 0 \cdot 0 + 2 \cdot \text{vol}(G) \cdot (\text{denominator}) \\ &= 4 \cdot \text{vol}(G) \cdot (\text{denominator}) \end{aligned}$$

In other words,

$$\text{denominator} = \frac{\sum_u \sum_v \deg(u) \deg(v) \cdot (f(u) - f(v))^2}{4 \cdot \text{vol}(G)}$$

Now, we use the same analysis in the main solution for the numerator:

$$\text{numerator} \geq \frac{1}{n} \cdot (f(v_{\max}) - f(v_{\min}))^2$$

And we bound the  $(f(u) - f(v))^2$  term in the denominator, which cancels with the numerator:

$$\text{denominator} \leq \frac{1}{4 \cdot \text{vol}(G)} \cdot \left( \sum_u \sum_v \deg(u) \deg(v) \right) \cdot (f(v_{\max}) - f(v_{\min}))^2$$

The double sum can be split into  $(\sum_u \deg(u) f(u)) \cdot (\sum_v \deg(v) f(v)) = (2 \cdot \text{vol}(G))^2$ . Therefore,

$$\frac{\text{numerator}}{\text{denominator}} \geq \frac{1/n}{(2 \cdot \text{vol}(G))^2 / (4 \cdot \text{vol}(G))} = \frac{1}{n \cdot \text{vol}(G)} = \Omega(n^{-3})$$

### 3 Problem 3

Consider the lazy random walk on a graph  $G$  and its mixing time  $T_{\text{mix}}(G, \varepsilon)$ .

Prove that there exists some distribution  $\mathbf{x}$  supported on a single vertex (i.e.  $\mathbf{x}_v = 1$  for some vertex  $v$ ,  $\mathbf{x}_u = 0$  for all other  $u$ ), such that it achieves the maximum in the definition of  $T_{\text{mix}}(G, \varepsilon)$ .

#### 3.1 My solution

For the sake of my sanity (to reduce symbols), write  $\mathbf{W} = \tilde{\mathbf{W}}^\top$ . We will never make use of the standard lazy walk matrix, so there is no confusion here. Also, the transpose doesn't matter because the matrix is symmetric.

Recall the total variation distance:

$$d_{\text{TV}}(\mathbf{x}, \mathbf{y}) = \frac{1}{2} |\mathbf{x} - \mathbf{y}|_1$$

We first make a few observations about  $d_{\text{TV}}$ .

First,  $d_{\text{TV}}$  is subadditive:

$$d_{\text{TV}}(\mathbf{a} + \mathbf{b}, \mathbf{c} + \mathbf{d}) \leq d_{\text{TV}}(\mathbf{a}, \mathbf{c}) + d_{\text{TV}}(\mathbf{b}, \mathbf{d})$$

The proof is by triangle inequality.

Second,  $d_{\text{TV}}$  scales linearly:

$$d_{\text{TV}}(c\mathbf{x}, c\mathbf{y}) = c \cdot d_{\text{TV}}(\mathbf{x}, \mathbf{y})$$

This is because the L1 norm also scales linearly.

We now go into the problem. Let  $\pi$  be the stationary distribution. For each vertex  $u$ , let  $\mathbf{e}_u$  be the basis vector that is 1 on the  $u$ -th component and 0 everywhere else. (This is a probability distribution supported only on  $u$ .) Define  $M$  to be the smallest natural number such that, for all vertex  $u$ ,

$$d_{\text{TV}}(\mathbf{W}^M \mathbf{e}_u, \pi) \leq \varepsilon.$$

In other words,  $M$  is the minimum number of steps such that every  $\mathbf{e}_u$  will become  $\varepsilon$ -close to  $\pi$ . Put another way, this is the definition of  $T_{\text{mix}}$  restricted to only distributions in the form  $\mathbf{e}_u$ .

Now let  $\mathbf{x}$  be any probability distribution, and write  $\mathbf{x} = \sum x_u \mathbf{e}_u$ . Note that  $x_u \geq 0$  everywhere and  $\sum_u x_u = 1$ . Also, let  $\pi$  be the stationary distribution. We will now write  $d_{\text{TV}}(\mathbf{W}^M \mathbf{x}, \pi)$  in terms of  $d_{\text{TV}}(\mathbf{W}^M \mathbf{e}_u, \pi)$ . This is pretty straightforward:

$$\begin{aligned} d_{\text{TV}}(\mathbf{W}^M \mathbf{x}, \pi) &= d_{\text{TV}}\left(\mathbf{W}^M \sum_u x_u \mathbf{e}_u, \pi\right) \\ &= d_{\text{TV}}\left(\mathbf{W}^M \sum_u x_u \mathbf{e}_u, \sum_u x_u \pi\right) && (\sum_u x_u = 1) \\ &\leq \sum_u d_{\text{TV}}(\mathbf{W}^M (x_u \mathbf{e}_u), x_u \pi) && (\text{subadditivity}) \\ &= \sum_u x_u \cdot d_{\text{TV}}(\mathbf{W}^M \mathbf{e}_u, \pi) && (\text{linear scaling}) \\ &\leq \sum_u x_u \cdot \varepsilon && (\text{definition of } M) \\ &= \varepsilon && (\sum_u x_u = 1) \end{aligned}$$

Therefore,  $\mathbf{x}$  also becomes  $\varepsilon$ -close to  $\pi$  within  $M$  steps. This is true for all  $\mathbf{x}$ , therefore  $T_{\text{mix}}(G, \varepsilon) \leq M$ . Clearly  $T_{\text{mix}}(G, \varepsilon) \geq M$  as well since one of the  $\mathbf{e}_u$  distributions needs that many steps, so this proves the claim.

### 3.2 Insight and discussion

The main insight is to note that we can basically handle the probability mass from each vertex separately, and add up all the masses together afterward. When I solved this problem, I literally visualized this by using paint colors: each vertex has some amount of paint of a distinct color, and the paint will spread throughout the graph.

Formalizing it looks scary due to the symbols, but ultimately, it's about using the subadditivity and linear scaling properties of  $d_{TV}$ . The subadditivity tells you that you can split  $\mathbf{x} = \sum_u x_u \mathbf{e}_u$ , and the total variation distance will just get worse; so if  $M$  steps are already enough for the  $\mathbf{e}_u$ 's, then  $M$  steps are also enough for  $\mathbf{x}$ . The linear scaling is due to the  $x_u$  coefficient there.

### 3.3 Grading criteria and common mistakes

There is no error notable enough to be mentioned for this problem. Well done!

## 4 Problem 4

Let  $G = (V, E)$  be a graph with adjacency matrix  $\mathbf{A}$ . Prove that

$$\lambda_{\max}(\mathbf{A}) \geq 2 \cdot \max_H \frac{|E(H)|}{|V(H)|}$$

where  $H$  is taken over all non-null subgraphs (not necessarily induced) of  $G$ .

### 4.1 My solution

Use the equation for largest eigenvalue in lecture 4 slide 14. Let  $R_{\mathbf{A}}$  be the Rayleigh quotient for the adjacency matrix  $\mathbf{A}$ :

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^{\top} \mathbf{A} \mathbf{x}}{\mathbf{x}^{\top} \mathbf{x}}$$

We now investigate what this Rayleigh quotient means.

Let  $\mathbf{x}$  be a 0-1 vector that is not all zero. This corresponds to a nonempty set  $S \subseteq V$ , specifically  $S = \{v : \mathbf{x}_v = 1\}$ .

The numerator  $\mathbf{x}^{\top} \mathbf{A} \mathbf{x}$  is the sum of entries in the form  $\mathbf{x}_u \mathbf{A}_{uv} \mathbf{x}_v$  over all  $u, v$ . Since  $\mathbf{A}$  is an adjacency matrix, it's 1 iff  $u, v$  are adjacent. Since  $\mathbf{x}$  is a 0-1 vector, this term is 1 iff  $u, v$  are adjacent and  $u, v$  are both in  $S$ , and 0 otherwise. Therefore, the numerator is  $2 \cdot |E(G[S])|$ . (There's a factor of 2 since an edge  $uv$  is counted twice, as  $u, v$  and as  $v, u$ .)

The denominator  $\mathbf{x}^{\top} \mathbf{x}$  is the sum of entries in the form  $\mathbf{x}_v^2$ . Since  $\mathbf{x}$  is a 0-1 vector, this term is 1 iff  $v \in S$ , and 0 otherwise. Therefore, the denominator is  $|V(G[S])|$ .

In other words, for a 0-1 vector  $\mathbf{x}$  represented by  $S$ ,

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^{\top} \mathbf{A} \mathbf{x}}{\mathbf{x}^{\top} \mathbf{x}} = \frac{2 \cdot |E(G[S])|}{|V(G[S])|}$$

Now,

$$\lambda_{\max}(\mathbf{A}) = \max_{\mathbf{x}} R_{\mathbf{A}}(\mathbf{x}).$$

Therefore, in particular,

$$\lambda_{\max}(\mathbf{A}) \geq R_{\mathbf{A}}(\mathbf{x})$$

over all 0-1 vectors  $\mathbf{x}$  that are not all zero.

Substituting in the observation,

$$\lambda_{\max}(\mathbf{A}) \geq 2 \cdot \frac{|E(G[S])|}{|V(G[S])|}$$

for all nonempty set  $S$ .

It just remains to generalize this to the  $H$  in the problem, because the problem says  $H$  is not necessarily induced. But this is trivial. If  $H$  has vertex set  $S$ , then  $|E(H)| \leq |E(G[S])|$ , so  $H$  will not have more edges than  $G[S]$ . So  $H$  will not give a better bound than  $G[S]$ . This proves the claim.

## 4.2 Insight and discussion

Because we work with the adjacency matrix  $\mathbf{A}$  instead of the normalized Laplacian, a whole ton of results from the lectures cannot be used. We don't have too many to work with; it's primarily the variational characterizations from the first half of lecture 4.

Considering this, we do know we have the bound for largest eigenvalue based on Rayleigh quotient:

$$\lambda_{\max}(\mathbf{A}) \geq R_{\mathbf{A}}(\mathbf{x})$$

for any  $\mathbf{x}$ . How can this relate to the problem?

Since the numerator  $|E(H)|$  counts edges, and the adjacency matrix  $\mathbf{A}$  involves edges, it's possible to add up entries of  $\mathbf{A}$  to count up edges. This may also lead to the 0-1 vector observation: an entry  $\mathbf{A}_{u,v}$  is counted iff  $u, v$  are present in the graph. To model whether a vertex is present or not, we may use a 0-1 vector for  $\mathbf{x}$ ; then  $\mathbf{A}_{u,v}$  is counted iff  $\mathbf{x}_u = \mathbf{x}_v = 1$ . And once you obtain this  $\mathbf{x}$  idea, it's pretty easy to show it works.

## 4.3 Grading criteria and common mistakes

### 4.3.1 Passable error: Assuming $H$ is induced

The problem statement says that  $H$  is not necessarily an induced subgraph. If you use the same  $\mathbf{x}$  idea as the main solution, then  $\mathbf{x}^\top \mathbf{A} \mathbf{x}$  counts all edges in  $G[S]$  – including edges not in  $H$ . If you did not account for this, it's strictly speaking an error.

It's very easy to fix though;  $H$  cannot have more edges than  $G[S]$ , so the fraction will just become smaller. So I don't deduct points for this. Just be careful and read the problem carefully.